2.4 GHz, 802.11ax WLAN Front-End Module

DESCRIPTION

The CB5337CX is a complete 2.4 GHz 802.11ax WLAN RF front-end module (FEM). The device provides all the functionality of a fully matched power amplifier (PA), power detector, low-noise amplifier (LNA), and single-pole, double-throw (SPDT) switch.

The CB5337CX provides a complete 2.4 GHz WLAN RF solution from the output of the transceiver to the antenna, and from the antenna to the input of the transceiver. The LNA increases the receive sensitivity of embedded solutions to improve range or to overcome the insertion loss of cellular filters (often included for mobile applications).

The CB5337CX also includes a transmitter power detector with 20 dB of dynamic range, and a digital enable control for transmitter power ramp on/off control. The device is provided in an ultra-compact, 16-pin 3 x 3 mm laminate package.

BLOCK DIAGRAM

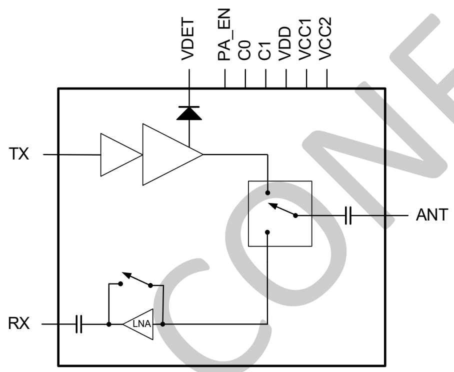


Figure 1. CB5337CX Block Diagram

FEATURES

- Integrated high-performance 2.4 GHz PA, harmonic filter, LNA with bypass, and T/R switch
- Fully matched input and output
- LNA with integrated bypass mode
- High-performance SPDT switch to support Wi-Fi functions
- Integrated positive slope power detector
- Transmit gain: 33 dB
- Receive gain: 14 dB
- Small 16-pin, 3 × 3 mm laminate package(MSL3, 260°C per JEDEC J-STD-020)

APPLICATIONS

- 802.11ax set-top boxes, networking, and personal computer systems
- PC cards, PCMCIA cards, mini-cards, and half mini-cards
- WLAN-enabled wireless video streaming systems

PIN-OUT DIAGRAM

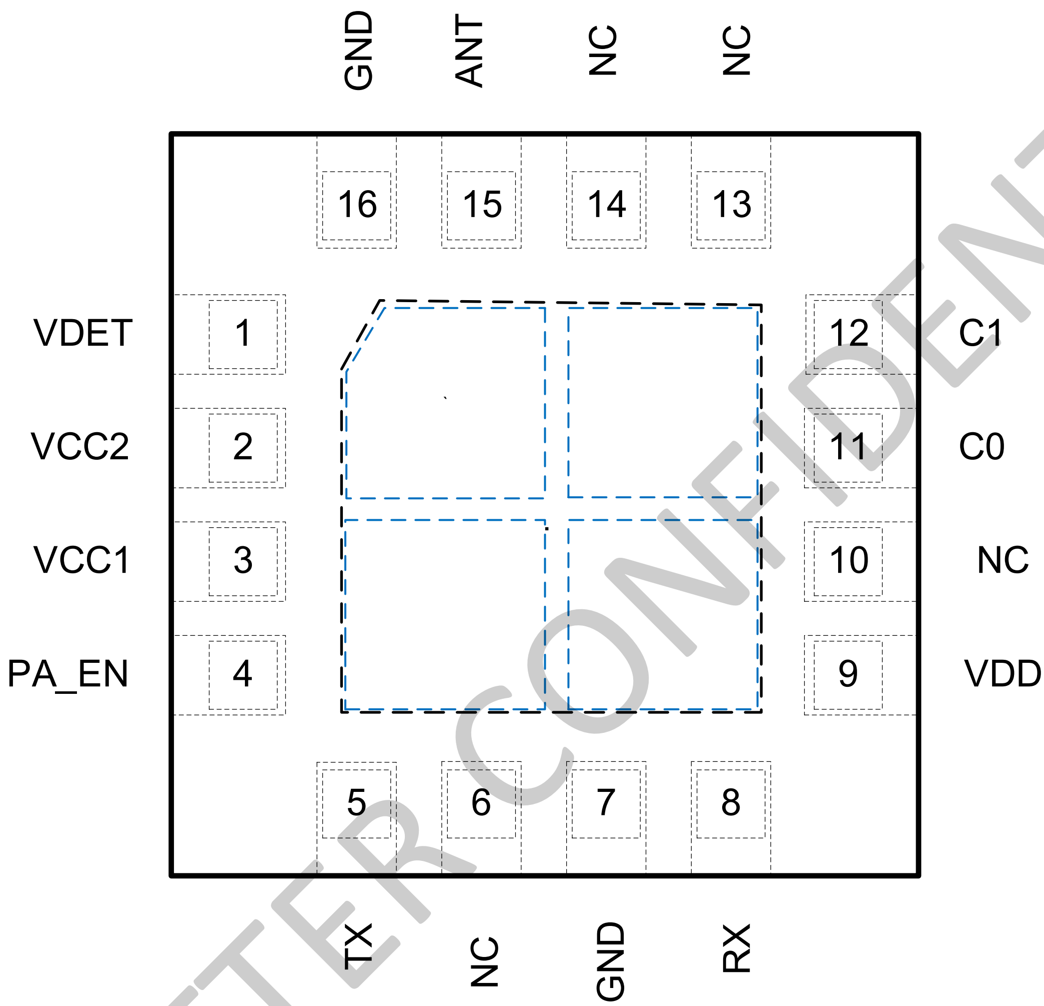


Figure 2. CB5337CX Pin out (Top View)

PIN ASSIGNMENTS

Pin	Name	Description	Pin	Name	Description
1	VDET	Power detector output voltage	9	VDD	RX supply voltage
2	VCC2	PA positive supply voltage	10	NC	No connection
3	VCC1	PA positive supply voltage	11	C0	Control input
4	PA_EN	PA enable control input	12	C1	Control input
5	TX	RF transmit PA input	13	NC	No connection
6	NC	No connection	14	NC	No connection
7	GND	Ground	15	ANT	Antenna
8	RX	RF receive LNA output	16	GND	Ground

CB5337CX ABSOLUTE MAXIMUM RATINGS

Parameters	Symbol	Minimum	Maximum	Units
Supply voltage	VCC1, VCC2, VDD	-0.3	+5.5	V
DC input on control pins (C0, C1 and PEN)	VIN	-0.3	3.6	V
Input power (50 Ω load)	PIN		+10	dBm
Storage temperature	Tst	-40	+150	°C
Junction temperature	TJ		+150	°C
Electrostatic discharge: Charged Device Model (CDM) Human Body Model (HBM)			 2000 1000	 V V

NOTE: Exposure to maximum rating conditions for extended periods may reduce device reliability. There is no damage to device with only one parameter set at the limit and all other parameters set at or below their nominal value. Exceeding any of the limits listed here may result in permanent damage to the device.

ESD HANDLING:	Although this device is designed to be as robust as possible, electrostatic discharge (ESD) can damage this device. This device must be protected at all times from ESD when handling or transporting. Static charges may easily produce potentials of several kilovolts on the human body or equipment, which can discharge without detection. Industry-standard ESD handling precautions should be used at all times.

RECOMMENDED OPERATING CONDITIONS

Parameters	Symbol	Minimum	Typical	Maximum	Units
Supply voltage	VCC1, VCC2, VDD		3.3	5	V
Control logic: High Low	 VIH VIL		 1.8 0		 V V
Quiescent current	ICQ, VCC1=VCC2= VDD=3.3V		160	185	mA
	ICQ, VCC1=VCC2= VDD=5V		180	200	mA
PA enable current	IENABLE		40	250	µA
LNA bias current (C0 = 3.3 V)	ICC_LNA		20	35	mA
LNA enable current (C0=C1 = 3.3 V)	ICC_LNA_BYP		40	850	µA
Operating frequency	f	2400		2500	MHz
Operating temperature	TC	-40	+25	+85	°C

ELECTRICAL SPECIFICATIONS

Parameters	Symbol	Test Condition	Min	Typ.	Max	Units
Transmit Mode：(VCC1 = VCC2 = VDD = 3.3 V, C0 = 0 V, C1 = PA_EN = 3.3 V, T_A = 25 °C, All Unused Ports Terminated with 50 Ω, Unless Otherwise Noted)						
Frequency	f	Main frequency band	2400		2500	MHz
Small signal gain	|S21|	@Pin= -25dBm		33.5		dB
Gain flatness	△S21	Over any 40 MHz bandwidth	-0.1		+0.1	dB
Output power	POUT	With -45 dB EVM source: DEVM = -40 dB, MCS11, HE40 DEVM = -35 dB, MCS11, VHT40 DEVM = -30 dB, MCS7, HT20 802.11n, MCS0, HT20 mask compliant		 +18 +20 +21 +26		 dBm dBm dBm dBm
Current consumption		Modulated signal: @ quiescent @ +20 dBm @ +24dBm @ +25 dBm Leakage, EN off		 160 250 344 375 850		 mA mA mA mA uA
Harmonics	2fo 3fo	POUT = +25 dBm, 1 Mbps, 802.11b POUT = +25 dBm, 1 Mbps, 802.11b		－50 －50		dBm/MHz dBm/MHz
Input return loss	|S11|	@ TX port		16		dB
Output return loss	|S22|	@ ANT port		20		dB
Saturation power	Psat	CW,@Pin= 5dBm		27.3		dBm
Power detector range	PDR	CW, measured @ ANT	5		29	dBm
Power detector output	VDET	POUT = No RF, measured into 1 MΩ POUT = +5 dBm, CW, measured into 1 MΩ POUT = +11 dBm, CW, measured into 1 MΩ POUT = +26 dBm, CW, measured into 1 MΩ		0.18 0.2 0.34 0.65		V V V V
Power detector slope	Slope	+11 dBm to +26 dBm		20.5		mV/dB
PA switching time		50% of PEN edge and 90/10% of final output power level			400	ns
Stability	STAB	CW, POUT = +26 dBm, 0.1 GHz to 20 GHz, load VSWR = 6:1	All non-harmonically related outputs less than -43 dBm/MHz			
Ruggedness	Ru	CW, PIN = +10 dBm, 0.1 GHz to 20 GHz, load VSWR = 10:1	No permanent damage or performance degradation			

Parameters	Symbol	Test Condition	Min	Typ.	Max	Units
Transmit Mode：(VCC1 = VCC2 = VDD = 5.0 V, C0 = 0 V, C1 = PA_EN = 3.3 V, T_A = 25 °C, All Unused Ports Terminated with 50 Ω, Unless Otherwise Noted)						
Frequency	f	Main frequency band	2400		2500	MHz
Small signal gain	|S21|	@Pin= -25dBm		34		dB
Gain flatness	△S21	Over any 40 MHz bandwidth	-0.1		+0.1	dB
Output power	POUT	With -45 dB EVM source: DEVM = -35 dB, MCS11, VHT40 DEVM = -30 dB, MCS7, HT20 802.11n, MCS0, HT20 mask compliant		 +22.5 +23 +26		 dBm dBm dBm
Current consumption		Modulated signal: @ quiescent @ +20 dBm @ +24dBm @ +25 dBm Leakage, EN off		 180 280 370 398 850		 mA mA mA mA uA
Harmonics	2fo 3fo	POUT = +25 dBm, 1 Mbps, 802.11b POUT = +25 dBm, 1 Mbps, 802.11b		-50 -50		dBm/MHz dBm/MHz
Input return loss	|S11|	@ TX port		16		dB
Output return loss	|S22|	@ ANT port		16		dB
Saturation power	Psat	@Pin= 5dBm		30		dBm
Power detector range	PDR	CW, measured @ ANT	5		29	dBm
Power detector output	VDET	POUT = No RF, measured into 1 MΩ POUT = +5 dBm, CW, measured into 1 MΩ POUT = +11 dBm, CW, measured into 1 MΩ POUT = +26 dBm, CW, measured into 1 MΩ		0.1 0.15 0.285 1.4		V V V V
Power detector slope	Slope	+11 dBm to +26 dBm		74		mV/dB
PA switching time		50% of PEN edge and 90/10% of final output power level			400	ns
Stability	STAB	CW, POUT = +26 dBm, 0.1 GHz to 20 GHz, load VSWR = 6:1	All non-harmonically related outputs less than -43 dBm/MHz			
Ruggedness	Ru	CW, PIN = +10 dBm, 0.1 GHz to 20 GHz, load VSWR = 10:1	No permanent damage or performance degradation			

Parameters	Symbol	Test Condition	Min	Typ.	Max	Units
Receive Mode: (VCC1 = VCC2 = VDD = 3.3 or 5V, C0 = 3.3 V, C1 = PA_EN = 0 V, T_A = 25 °C, All Unused Ports Terminated with 50 Ω, Unless Otherwise Noted)						
Frequency	f	Main frequency band	2400		2500	MHz
Small signal gain	|S21|	LNA active (C0 = 3.3 V, C1 = PEN = 0 V) LNA bypass (C0 = C1 = 3.3 V, PEN = 0 V)		+15 -4		dB dB
Noise figure	NF	From ANT to RX pins		1.6	1.8	dB
Input return loss	|S11|	LNA active (C0 = 3.3 V, C1 = PEN = 0 V) LNA bypass (C0 = C1 = 3.3 V, PEN = 0 V)		10 13		dB dB
Output return loss	|S22|	LNA active (C0 = 3.3 V, C1 = PEN = 0 V) LNA bypass (C0 = C1 = 3.3 V, PEN = 0 V)		6.5 13		dB
ThirdP order input intercept point	IIP3	LNA active (C0 = 3.3 V, C1 = PEN = 0 V) LNA bypass (C0 = C1 = 3.3 V, PEN = 0 V)		+12 +30		dBm dBm
Switch isolation - ANT to LNA output	ISO	In TX mode		31		dB
Switching time	tSW	LNA ↔ bypass RX ↔ TX			200 500	ns ns

NOTE:
1. Performance is guaranteed only under the conditions listed in this table.
2. Guaranteed by characterization.

LOGIC TRUTH TABLE

Mode	State	C0(Pin 11)	C1(Pin 12)	PA_EN (Pin 4)
All off	1	0	0	0
WLAN receive LNA	2	1	0	0
WLAN receive bypass	3	1	1	0
WLAN transmit, high linearity	4	0	1	1
All other states	Not supported			

EVALUATION BOARD SCHEMATIC

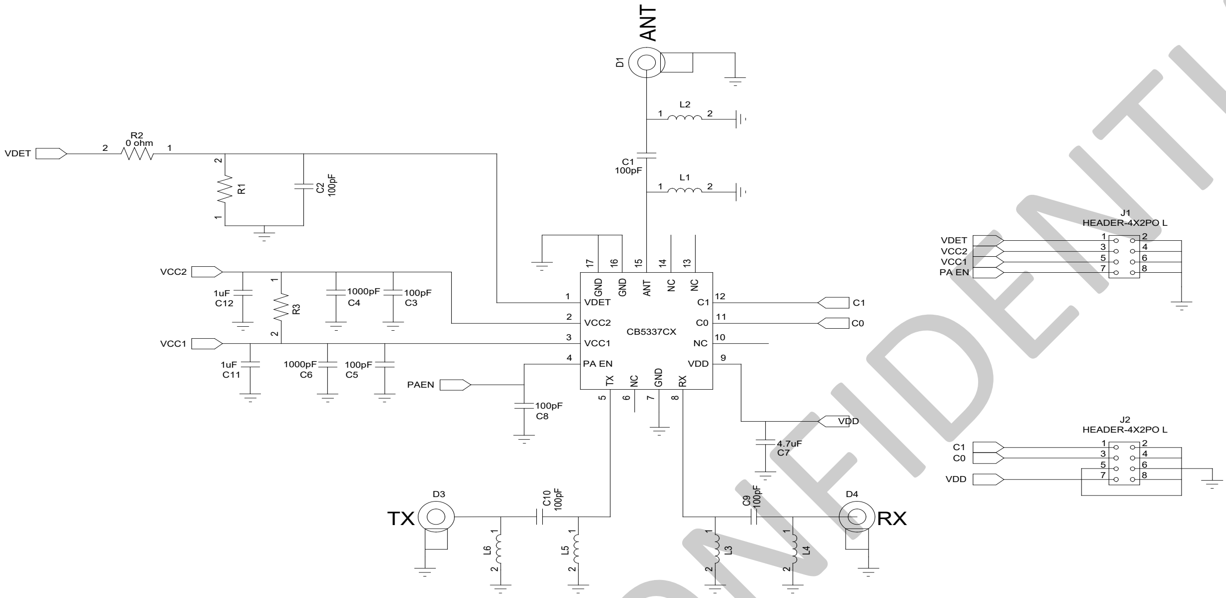


Figure 3. CB5337CX Evaluation Board Schematic

EVALUATION BOARD ASSEMBLY DRAWING

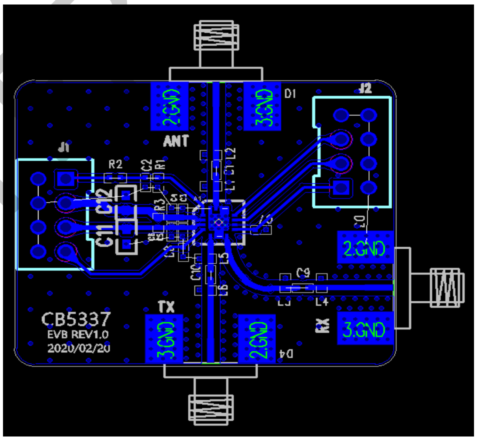
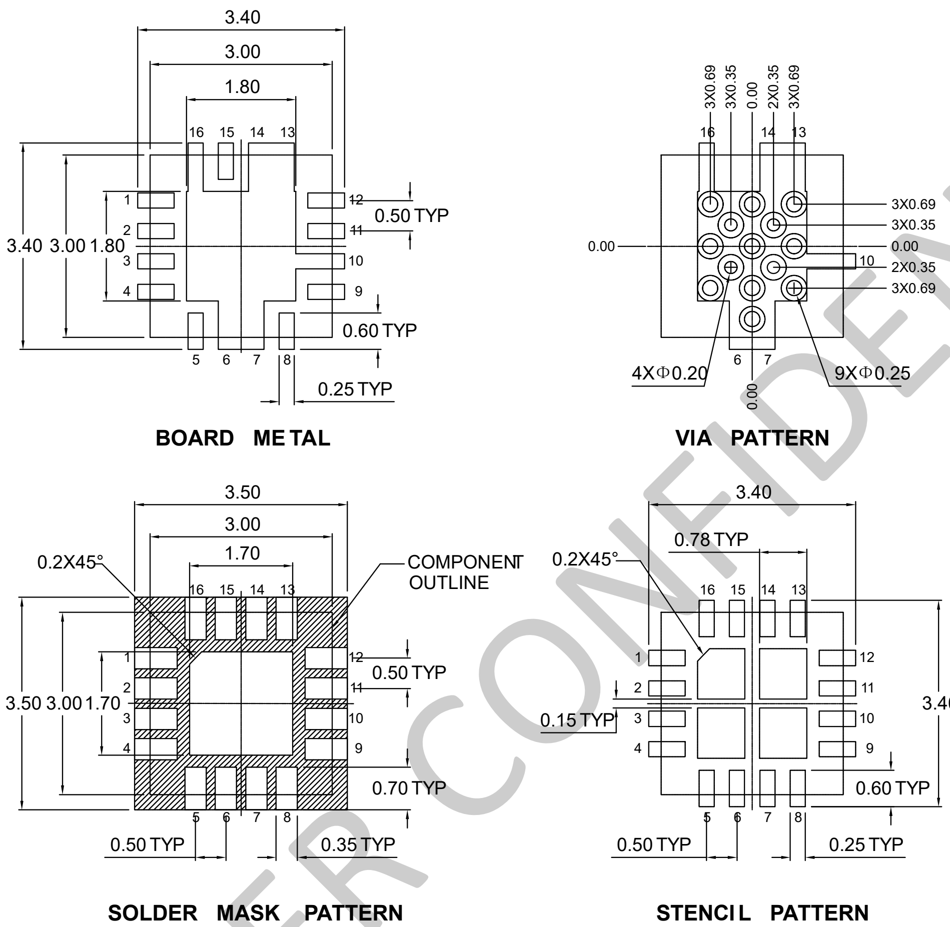


Figure 4. CB5337CX Evaluation Board Assembly Drawing

BILL OF MATERIALS

Component	Value	Size	Vendor	Part Number
C11, C12	1 μF	0805	Murata	
C4, C6	1000 pF	0402	Murata	
C1, C2, C3, C5,, C8, C9, C10	100 pF	0402	Murata	
C7	4.7uF	0402	Murata	
R2	0 Ω	0402	Uni-Royal	
L1~L6, R1, R3	DNI			

PCB LAND PATTERN



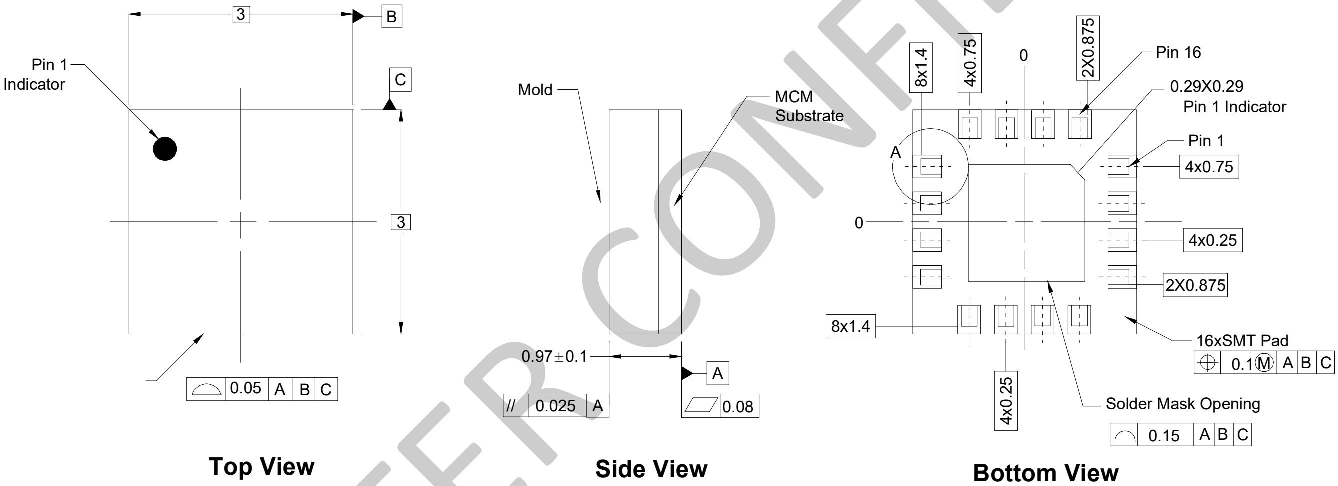
NOTE: THERMAL VIAS SHOULD BE RESIN FILLED AND CAPPED IN ACCORDANCE WITH IPC-4761 TYPE VII VIAS. 30-35UM Cu THICKNESS IS RECOMMENDED.

Figure 5. CB5337CX PCB Layout Footprint

TYPICAL PART MARKING

Figure 6. Typical Part Marking for the CB5337CX

PACKAGE DIMENSIONS (All Dimensions in mm):



Notes:
1. Dimensions are in millimeters.
2. Dimensions and tolerances according to ASME Y14.5M-1994
3. See applicable bonding diagram and device assembly drawing for die and component placement.
4. Pad definitions per details on drawing.
5. PCB Type 4L PPG TEV MCM(100).

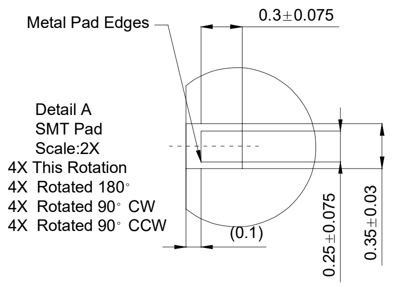


Figure 7. CB5337CX Package Dimension

CONTACT INFORMATION

For the latest specifications, additional product information, worldwide sales and distribution locations:

Web: www.chipbetter.com
Tel: 0755-26654180

DISCLAIMERS

ChipBetter reserves the right to make changes without further notice to specifications and product descriptions in this document to improve reliability, function or design. ChipBetter does not assume any liability arising out of the application or use of information or product described in this document. Neither does ChipBetter convey any license under its intellectual property rights nor licenses to any of circuits described in this document to any third party.
The information in this document is believed to be accurate and reliable and is provided on an "as is" basis, without any express or implied warranty. Any information given in this document does not constitute any warranty of merchantability or fitness for a particular use. The operation of this product is subject to the user's implementation and design practices. ChipBetter products are not designed or intended for use in life support equipment, devices or systems, or other critical applications, and are not authorized or warranted for such use.